

DAY 06: AWS VPC, SUBNETS, GATEWAYS, ROUTE TABLES, PEERING & CLOUDWATCH

Cloud Computing & DevOps Master Curriculum

1. Amazon VPC (Virtual Private Cloud)

AWS VPC lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You control IP ranges (CIDR blocks), subnets, and route tables.

2. Core Networking Components

Component	Description
Public Subnet	Subnet associated with a route table containing a route to an Internet Gateway (IGW). Instances receive public IPs.
Private Subnet	Subnet with no direct route to IGW. Used for backend databases, microservices, and internal systems.
Internet Gateway (IGW)	VPC component that allows communication between instances in your public subnets and the internet.
NAT Gateway	Network Address Translation service allowing instances in private subnets to connect to the internet while preventing internet-initiated connections.
Route Tables	Set of rules (routes) used to determine where network traffic from your subnet or gateway is directed.

3. Advanced VPC Connectivity & CloudWatch

VPC Peering: Networking connection between two VPCs that enables you to route traffic between them using private IPv4/IPv6 addresses directly.

Transit Gateway: A network transit hub that connects VPCs and on-premises networks through a central router, simplifying complex hub-and-spoke topologies.

Amazon CloudWatch: Monitoring and observability service that collects logs, metrics, and events from EC2, VPC, Lambda, and sets alarms to trigger automated notifications or autoscaling actions.

KEY TAKEAWAYS & SUMMARY

- ✓ Public subnets route traffic to Internet Gateway; private subnets use NAT Gateway for outbound connectivity.
- ✓ VPC Peering connects 2 VPCs point-to-point; Transit Gateway simplifies multi-VPC hub-and-spoke networks.
- ✓ CloudWatch provides full-stack visibility across AWS infrastructure metrics and log streams.